

HASSAN KHAN

Software Engineering Student | Full-Stack & AI Development

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Portfolio: hassan-khan-5535.github.io/my-portfolio/

PROFESSIONAL SUMMARY

Software Engineering student at COMSATS University Islamabad, Abbottabad Campus, with a 3.4 CGPA and a strong foundation in programming, data structures, databases, artificial intelligence, and software engineering. Developed academic projects spanning AI, computer vision/OCR, database applications, and JavaFX software. Proficient in C++, C, Java, Python, JavaScript, HTML, CSS, SQL, Git, and GitHub, with hands-on experience integrating Gemini and Grok APIs. Currently expanding into MERN and modern full-stack development.

EDUCATION

Bachelor of Science in Software Engineering - COMSATS University Islamabad, Abbottabad Campus

2024-2028 | CGPA: 3.4/4.0

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Artificial Intelligence, Operating Systems, Computer Networks, Programming Fundamentals, Software Engineering, Professional Practices, Discrete Structures

TECHNICAL SKILLS

Languages: C, C++, Java, Python, JavaScript, SQL

Web / Full-Stack: HTML5, CSS3, JavaScript, full-stack application development; currently learning MERN, React, Node.js, Express.js, MongoDB

AI / APIs: Google Gemini Vision API, xAI/Grok API, LLM/API integration, AI-assisted application development

Computer Vision / Data: OpenCV, NumPy, pyzbar, EasyOCR, pandas, openpyxl, Streamlit

Databases: MySQL, SQL, JDBC

Tools: Git, GitHub, VS Code, Python dotenv

Core Concepts: OOP, DSA, MVC, REST/API concepts, database connectivity, software engineering fundamentals

SELECTED PROJECTS

Automated Quiz Scanner & Grading System

Python, OpenCV, NumPy, pyzbar, Google Gemini Vision API, EasyOCR, pandas, openpyxl, Streamlit

- Built an AI-powered quiz scanning and grading application that processes bubble sheets, decodes QR-based answer keys, and extracts handwritten student information.
- Integrated Gemini 2.5 Flash for handwritten text recognition with EasyOCR as a fallback, and added batch processing with CSV/Excel report export.
- Developed a browser-based interface in Streamlit and managed configuration securely with python-dotenv.

GitHub: <https://github.com/Hassan-khan-5535/quiz-scanner-and-grading-system>

Othello / Reversi - Two-Player AI Game

Java, JavaFX, CSS, OOP, MVC, AI/heuristic algorithms

- Developed a fully functional Othello/Reversi game with Player vs Player and Player vs Computer modes.
- Implemented move validation across eight directions, turn handling, score tracking, undo using a stack, and an AI opponent using a board-position heuristic strategy.
- Applied MVC architecture to separate model, view, and controller responsibilities and built a custom JavaFX/CSS interface.

GitHub: <https://github.com/Hassan-khan-5535/Othello-Reversi>

EchoAgent - Voice-Native AI Assistant

Python, FastAPI, React, Claude API, faster-whisper, SQLite, Tailwind CSS

- Developed a bilingual (English & Urdu) voice-native agentic AI assistant capable of reasoning and executing real-world tool calls via Anthropic Claude.
- Engineered an ultra-low latency voice pipeline using CPU-optimized faster-whisper (STT) and native edge-tts (TTS).
- Built a real-time web UI using React and Vite, featuring visible reasoning traces and cross-session memory backed by SQLite.

GitHub: <https://github.com/Hassan-khan-5535/echoagent>

CERTIFICATIONS

- Web Development Internship & Training Program - Arch Technologies | May 2026
- Agile Project Management - HP LIFE | May 2026
- Introduction to Cybersecurity Awareness - HP LIFE | February 2026
- Introduction to Modern AI - Cisco Networking Academy | February 2026

ADDITIONAL EXPERIENCE & INTERESTS

- Hands-on experience integrating external AI/LLM APIs, including Google Gemini and xAI/Grok, into software projects.
- Actively developing skills in MERN stack development and modern AI-powered applications.
- Career interests: Software Engineering, Full-Stack/MERN Development, AI-powered Applications, API Integration, and Freelance Development.

LANGUAGES & CAREER TARGET

Languages: English - Professional | Urdu - Professional

Target: Software Engineering Internship | Full-Stack Developer Internship | MERN Developer | Freelance Developer | Remote